Thomas O. Merritt

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Profiles: https://thomasmerritt.github.io | LinkedIn: in/thomas-merritt | Github: *ThomasMerritt*
SKILLS: Python (Asyncio), JavaScript, SQL; React, NextJS, REST APIs, Selenium, GenAI Integration (Claude/GPT SDKs), RAG, MCP, Azure OpenAI (Prism Gateway), AWS, Anthropic

RELEVANT EXPERIENCE

NBCUniversal | Software Engineering & AI Solutions | Los Angeles, CA ***Apr 2026 - Present***

Frontend Architecture & RAG Integration

- Led migration of an executive-facing Finance AI platform from a Streamlit prototype to a high-performance Next.js ecosystem with robust client-side state management.
- Powers real-time data streaming and conversational analytics for C-Suite executives and global finance leaders, scaling the system to reliably process and validate 10,000+ budget records securely.
- Integrated Next.js UI components with a production RAG backend, building the core chat interface, sidebar virtualization, live streaming pipelines, and custom answer animations.

AI Automation, Validation & Data Engineering

- Built an automated "LLM-as-a-Judge" pipeline using Claude/GPT APIs to ingest system semantic layers and business rules, fully automating compliant markdown documentation.
- Engineered a concurrent async testing pipeline within the platform's Admin Panel to batch-evaluate hundreds of prompt strings; refactored workflows to bring execution runtimes from ~1 hour → minutes.
- Developed SQL sampling logic for a development dataset, capping data by cycle and month while preserving sales-versus-finance distributions across full calendar years.

Backend, Reliability & Security Hardening

- Updated conversation APIs, metadata schemas, and guardrail-safe SQL persistence layers to harden query semantics and maintain absolute frontend/backend state synchronization.
- Remediated enterprise security review vulnerabilities, hardening the application against authentication bypass, restricting Swagger access, and configuring secure logging profiles.

Agilisium | Software Engineer Intern | Los Angeles, CA **Jan *2026 - Feb 2026***

- Monitored high-traffic AWS systems via Datadog and CloudWatch to triage ECS health failures, API spikes, and DynamoDB/DAX performance anomalies.

Telecom Infrastructure Partners | Software Engineer Intern | London, UK ***Jun 2024 - Sep 2024***

- Automated European cell tower data extraction using a Selenium web scraper, eliminating 200+ hours of manual effort annually and accelerating data availability for corporate decision-making.
- Designed and wireframed a full-stack geographic analysis application leveraging Azure Maps and Microsoft 365 CRM, integrating real-time data pipelines and interactive map controls to maximize asset pricing accuracy and operational efficiency.

TRAPNSTUDIO | Software Engineer Intern | Riverside, CA ***Feb 2024 - May 2024***

- Contributed to the development of Vibecue, a DJ-driven music software designed to manage and control music queues in public venues; gained exposure to software engineering workflow and environments

PROJECTS

ON-DEVICE GENAI IMAGE PIPELINE | Personal Project

- Built a FLUX-targeted character concept engine in JavaScript — a recursive {wildcard} prompt grammar with weighted constraint filtering across 130 character profiles.

HOLLYWHATIF | Personal Project

- Built a full-stack web application using Flask and Apache Spark to predict the impact of cast changes on IMDb ratings through real-time linear regression analysis.

EDUCATION

University of California Riverside | Riverside, CA ***2024***

B.S. in Computer Science | GPA: 3.64/4.0

Certifications: AWS Certified Developer – Associate (DVA-C02)